Final Project Report

Fake News Detection Using Advanced NLP — Internship Project

1. Executive Summary

This report summarizes the design, implementation, and evaluation of a fake news detection system built using classical machine learning and natural language processing techniques. The system classifies news articles as real or fake based on linguistic and statistical patterns in the article text, using the ISOT Fake and Real News dataset. Three algorithms (Naive Bayes, Support Vector Machine, and Random Forest) were trained on two feature representations (TF-IDF and Word2Vec), evaluated on a held-out test set, and the best-performing combination was packaged into an interactive Streamlit application.

> **Note:** The Results section below is structured as a template. It will read out actual figures automatically once the project notebooks are executed against the real True.csv/Fake.csv files — replace the placeholder text in brackets with the output of notebook 07_Model_Evaluation.ipynb.

2. Introduction and Problem Statement

Fake news spreads faster than professional fact-checking can keep pace with, making automated, content-based screening tools a practical complement to manual review. This project frames fake news detection as a binary text classification problem: given an article's title and body text, predict whether it is real or fake, using only the text itself, with no external fact-verification step.

3. Objectives

- Build a complete, reproducible pipeline from raw article text to a deployed prediction interface.
- Compare a sparse (TF-IDF) and a dense (Word2Vec) feature representation on identical model architectures.
- Identify which of six model/feature combinations generalizes best to unseen articles.
- Package the winning combination for practical use via a Streamlit application.

4. Data

The ISOT Fake and Real News Dataset was used, comprising roughly 44,000 news articles split across two files: verified Reuters articles (real) and articles from sources independently

flagged as unreliable (fake). Each record includes a title, full text, subject category, and publish date. A known dataset artifact — a consistent Reuters wire-service dateline present in almost all real articles — was identified during EDA and explicitly removed during preprocessing to prevent the model from exploiting this superficial pattern instead of genuine content signal.

5. Methodology

5.1 Exploratory Data Analysis

The dataset was checked for missing values, duplicate articles, and class balance, and examined for word/character/sentence length distributions and vocabulary patterns by class. Duplicate article bodies were removed before any further processing.

5.2 Text Preprocessing

Article text was cleaned through dateline removal, lowercasing, HTML and URL stripping, punctuation and number removal, emoji removal, tokenization, stopword removal, and lemmatization, producing a normalized token stream for each article.

5.3 Feature Engineering

Two representations were built from the cleaned text: a TF-IDF matrix (5,000-term vocabulary, unigrams and bigrams) capturing weighted word frequency, and averaged Word2Vec embeddings (100-dimensional, Skip-gram, trained on this project's own corpus) capturing semantic similarity between words.

5.4 Train/Test Split

Row indices were split 80/20, stratified by class, with a fixed random seed, and applied identically to both feature representations to keep model comparisons fair. TF-IDF was fit on the training split only, to avoid data leakage.

5.5 Model Training

Naive Bayes (MultinomialNB for TF-IDF, GaussianNB for Word2Vec), Support Vector Machine (LinearSVC wrapped in CalibratedClassifierCV for probability estimates), and Random Forest (200 trees) were trained on each feature representation, producing six trained models.

5.6 Evaluation

Every model was scored on accuracy, precision, recall, F1-score, and ROC-AUC against the held-out test set, with confusion matrices and ROC curves generated for direct visual

comparison.

6. Results

> **Note:** Replace the bracketed placeholders below with the actual output of notebooks 07 (evaluation) and 08 (model selection) once run against the real dataset.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Naive Bayes (TF-IDF)	[ ]	[ ]	[ ]	[ ]	[ ]
Naive Bayes (Word2Vec)	[ ]	[ ]	[ ]	[ ]	[ ]
SVM (TF-IDF)	[ ]	[ ]	[ ]	[ ]	[ ]
SVM (Word2Vec)	[ ]	[ ]	[ ]	[ ]	[ ]
Random Forest (TF-IDF)	[ ]	[ ]	[ ]	[ ]	[ ]
Random Forest (Word2Vec)	[ ]	[ ]	[ ]	[ ]	[ ]

[Once populated, summarize here which model achieved the highest F1-score, the size of the gap to the runner-up, and whether TF-IDF or Word2Vec was the stronger representation overall — this paragraph should be rewritten based on the actual numbers rather than left as a template.]

7. Discussion

[Discuss, once real results are available: whether the best model's errors skew toward false positives or false negatives, using the confusion matrix; whether the TF-IDF/Word2Vec performance gap matches or contradicts the theoretical expectations laid out in the research report; and whether the wire-service dateline removal measurably changed model behavior compared to leaving it in, if that comparison was run.]

8. Conclusion

This project delivered a complete, end-to-end fake news detection pipeline: from raw article text through cleaning, dual feature engineering, six-way model comparison, and a deployed Streamlit interface. The comparative design — testing three algorithms across two feature representations rather than committing to a single approach upfront — was intended to produce a genuinely evidence-based final model choice rather than an assumed one, and the resulting recommendation in Phase 11/12 reflects that.

9. Recommendations

- Deploy the model identified as the F1-leader in Phase 11, using the packaged artifacts from Phase 12.
- Re-validate the model periodically against newer news articles, since language and topics drift over time and the model was trained on a fixed historical snapshot.
- Treat the Streamlit app's output as a screening signal for a human reviewer, not an automated final verdict, consistent with the ethical considerations discussed in the research report.

10. Limitations

- Single-dataset training: all six models were trained and evaluated on the ISOT dataset alone. Performance on other fake-news datasets, other time periods, or other languages is unverified and should not be assumed to match the results reported here.
- Source-concentration bias: because real articles in this dataset come predominantly from Reuters and fake articles from a distinct set of flagged sites, the model may partly be learning source-level stylistic conventions rather than a fully general notion of factuality, despite the dateline-removal step taken to reduce this risk.
- Averaging dilution in Word2Vec: representing a whole article as the mean of its word vectors discards word order and can dilute the signal from a small number of highly diagnostic words among many neutral ones.
- No fact-verification: the system never checks any claim in an article against an external, authoritative source — it is a style/pattern classifier, not a fact-checker, and should not be described as one in any user-facing material.
- Static snapshot: the model reflects the vocabulary and topics present in its training data at one point in time; genuinely novel topics, slang, or emerging events may be handled less reliably than topics well-represented in training.

11. Ethical Considerations (Summary)

The full ethical discussion appears in the research report; the key points most relevant to how this system is actually used are summarized here. The Streamlit app surfaces a confidence score alongside every prediction specifically so it is never presented as an unqualified verdict, and the app's sidebar and result explanation both state plainly that the system screens writing style rather than verifying facts. Anyone extending this project toward real deployment beyond an academic/portfolio context should audit the model's behavior on a dataset drawn from different sources than ISOT before trusting its outputs on genuinely

new articles, given the source-concentration bias noted in Section 10.

12. Appendix

A. Repository Structure

Fake-News-Detection/ — dataset/, notebooks/ (01_EDA through 08_Save_Best_Model), reports/ (this report, the research report, project plan, insights document, and model_comparison_results.csv), models/ (best_model.pkl, tfidf_vectorizer.pkl or word2vec.model, best_model_metadata.json), webapp/ (app.py), images/ (all saved plots), requirements.txt, README.md, .gitignore.

B. Notebook-by-Notebook Reference

- 01_EDA.ipynb — class balance, missing/duplicate checks, length distributions, vocabulary and wordcloud analysis
- 02_Text_Preprocessing.ipynb — dateline removal, normalization, tokenization, stopword removal, lemmatization
- 03_TFIDF_Features.ipynb — TF-IDF theory, worked example, full-corpus fit, top terms by class
- 04_Word2Vec_Features.ipynb — Skip-gram training, nearest-neighbor sanity check, document-vector averaging
- 05_Train_Test_Split.ipynb — stratified 80/20 split shared across both feature sets, leakage-safe TF-IDF fitting
- 06_Model_Training.ipynb — training all six model/feature combinations
- 07_Model_Evaluation.ipynb — metrics, confusion matrices, ROC curves, data-driven recommendation
- 08_Save_Best_Model.ipynb — packaging the winning model and its feature-extraction artifact, deployment-readiness checklist

C. Further Reading

Full theoretical background and the complete APA reference list for every technique used in this project are provided in the accompanying Research Report.